

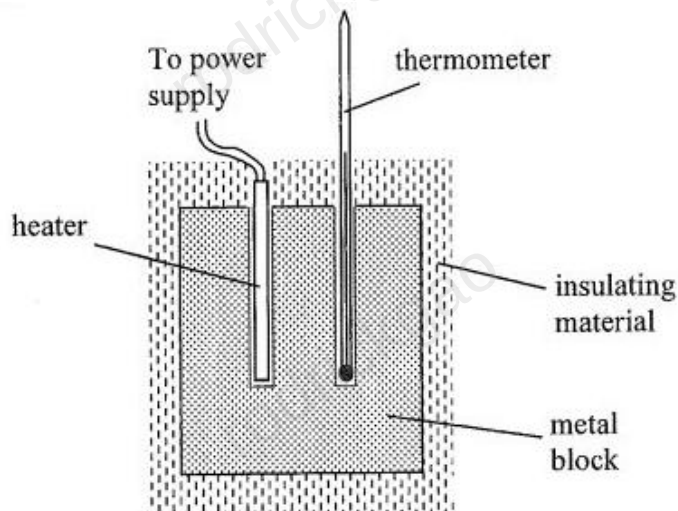
2014 HKDSE Physics

Paper 1B Structured Questions

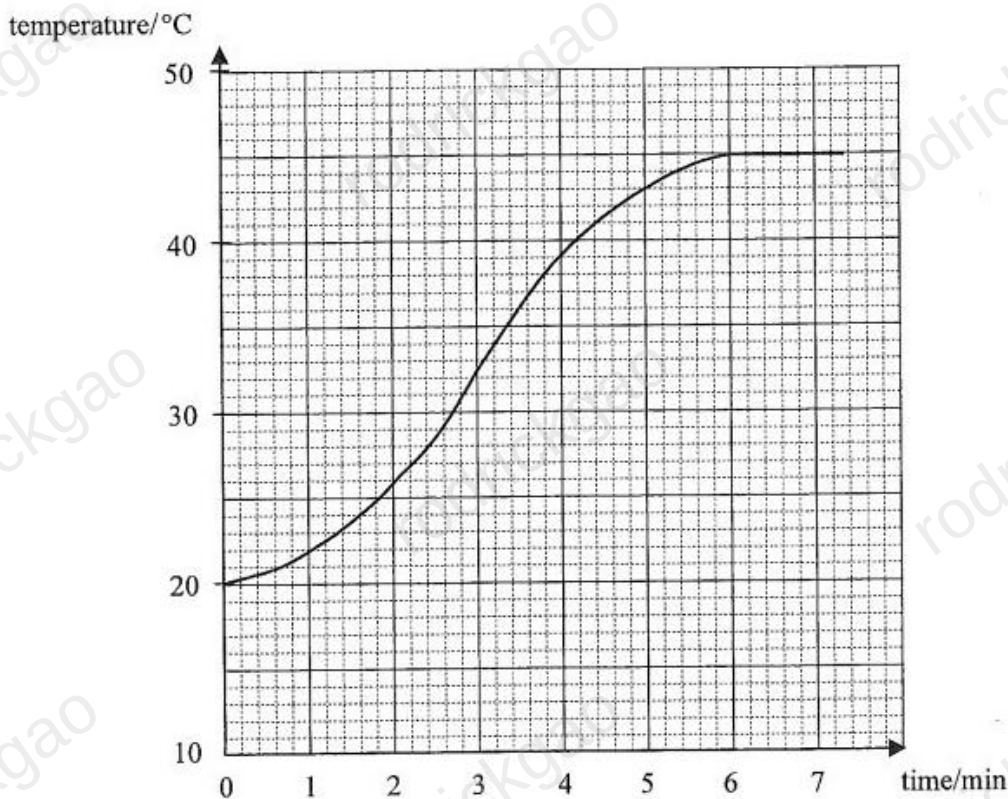
Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Question 1

Figure 1.1 shows an experiment to find the specific heat capacity of a metal. An insulated metal block contains a heater and a thermometer. The heater is switched on at 20°C and off when the block reaches 43°C .



The graph shows the block temperature against time.



- (a) Use the graph to find how long the heater is switched on. (1 mark)
- (b) Explain why the block temperature continues to rise for a while after switch-off. (1 mark)
- (c) The metal mass is 0.80 kg, heater voltage 12 V and current 4.0 A.
- (c)(i) Using the maximum temperature rise, calculate the experimental specific heat capacity. (2 marks)
- (c)(ii) Is this result equal to, higher or lower than the actual value? Explain. (2 marks)
- (d) Explain why this method is unsuitable for a glass block. (1 mark)

Official marking scheme

Solution	Marks	Remarks
1. (a) 5 min. (or 300 s)	1A	
	1	
(b) When the heater is switched off, its temperature is higher than the metal.	1A	
Or Heat continues to conduct to the metal until the same temperature is attained.	1A	
Or It takes time for them to attain the same temperature.	1A	
	1	
(c) (i) $mc\Delta T = IVt$ $(0.80)c(45 - 20) = (4.0)(12)(5 \times 60)$ $c = 720 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$	1M 1A	
	2	
(ii) Experimental result is higher than the actual value. Not all the energy supplied by the heater goes to the metal.	1A 1A	
Or Some energy is absorbed by the heater itself and / or by the thermometer.	1A	
Or Heat loss to the surroundings.	1A	
	2	
(d) Glass is not a good conductor of heat.	1A	
Or Prolonged time is required for the whole glass block to attain uniform temperature.	1A	
Or No insulating material is perfect, so heat loss is appreciable.	1A	
	1	

(a) Read the time at 43°C, when the supply is cut: 5 min. The later plateau does not mark the switch-off time.

(b) Switching off stops electrical input, but does not instantly cool the heater. Its stored thermal energy continues to warm the metal.

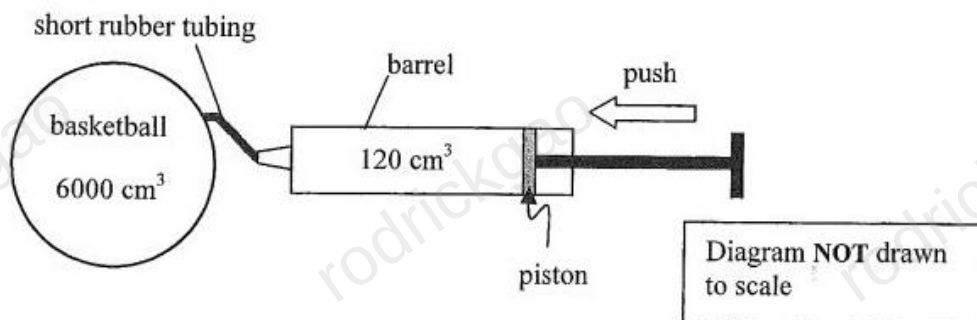
(c)(i) Energy supplied = $IVt = 4.0 \times 12 \times 300 = 14\,400 \text{ J}$. The maximum rise is $45 - 20 = 25^{\circ}\text{C}$, not 23°C . Therefore $c = 14\,400 / (0.80 \times 25) = 720 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

(c)(ii) This calculation attributes all electrical energy to the metal. Since some energy goes elsewhere, the numerator is too large for the observed metal temperature rise and c is overestimated.

(d) Low thermal conductivity produces substantial temperature differences within glass. Waiting for a uniform temperature allows more heat loss, undermining this method.

Question 2

* A basketball is connected to an air pump by short rubber tubing. Each inward stroke pumps all 120 cm^3 of air originally in the barrel at atmospheric pressure and room temperature into the ball. Valves prevent backflow when the piston is pulled out.



Initially the basketball contains 6000 cm^3 of air at atmospheric pressure 100 kPa . It must reach 156 kPa for an official match. Assume constant room temperature throughout.

(a)(i) Show that 3360 cm^3 of air, measured at atmospheric pressure, must be added. The ball volume stays at 6000 cm^3 . (3 marks)

(a)(ii) Estimate the minimum number of strokes required. (1 mark)

(b) Use the kinetic theory of an ideal gas to explain the pressure increase when air is pumped into the ball. (2 marks)

Official marking scheme

Solution	Marks	Remarks
2. (a) (i) V_0 = total volume of air required at P_0 , including the residual air in the ball $P_1 V_1 = P_0 V_0$ $(156 \text{ kPa}) (6000 \text{ cm}^3) = (100 \text{ kPa}) (V_0)$ $V_0 = 9360 \text{ cm}^3$ $\therefore$ volume of air = V_0 – volume of the basketball $= 9360 \text{ cm}^3 - 6000 \text{ cm}^3$ $= 3360 \text{ cm}^3$	1M 1A 1M	
<i>Alternatively:</i> $\therefore n = \frac{PV}{RT}$ $\frac{156V}{RT} - \frac{100V}{RT} = n'$ $\therefore V' = \frac{56V}{RT} \cdot \frac{RT}{100} = 3360 \text{ cm}^3$	1M 1M + 1A	
	3	
(ii) Number of strokes required = $3360 \text{ cm}^3 \div 120 \text{ cm}^3$ $= 28$	1A	
	1	
(b) Using kinetic theory, the pressure p is given by $pV = \frac{Nm\overline{c^2}}{3} = \frac{2N}{3} \cdot \frac{mc^2}{2}$ As the volume and the temperature ($\propto$ kinetic energy of the air molecules) remain unchanged, the increase in pressure is due to the increase of number of air molecules hitting the wall of the container per unit time. Or Frequency of collision with the wall of container increases.	1A 1A 1A	
	2	

(a)(i) Apply Boyle's law to the fixed collection of all air that will end up in the ball. At 100 kPa this collection occupies V_0 ; at 156 kPa it occupies 6000 cm^3 . Since both pressures and volumes use matching units, their ratios may be calculated in kPa and cm^3 .

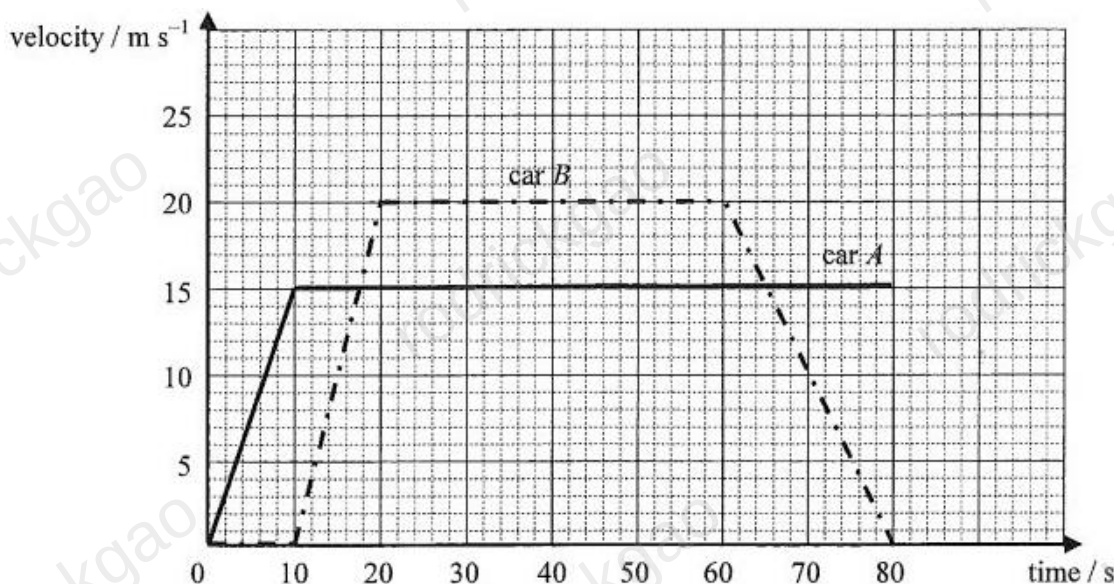
The 9360 cm^3 includes the initial 6000 cm^3 ; only the difference must be supplied by the pump. Equivalently,
 $V_{\text{added}} = [(156-100)/100] \times 6000 = 3360 \text{ cm}^3$.

(a)(ii) Each ideal stroke supplies 120 cm^3 measured under the same atmospheric conditions, so 28 complete strokes suffice.

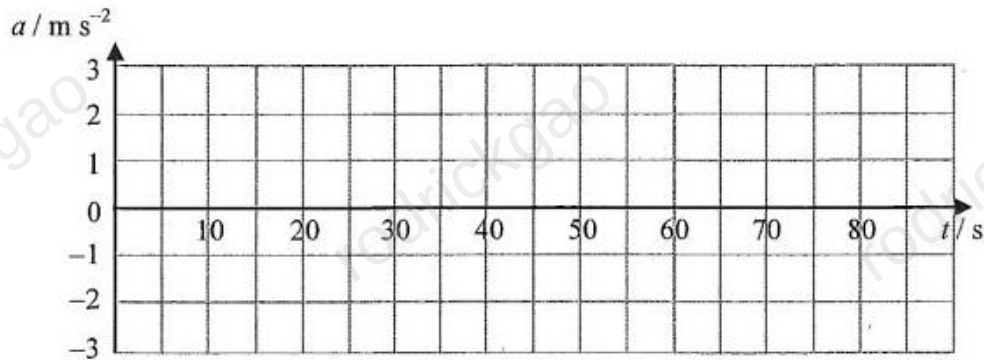
(b) Constant temperature means no rise in average molecular kinetic energy. Adding molecules at fixed volume increases the rate of momentum transfer to each unit area of wall, increasing pressure. Do not claim that molecules move faster here.

Question 3

Cars A and B start from the same position on the same straight horizontal road. Their velocities vary with time as shown.

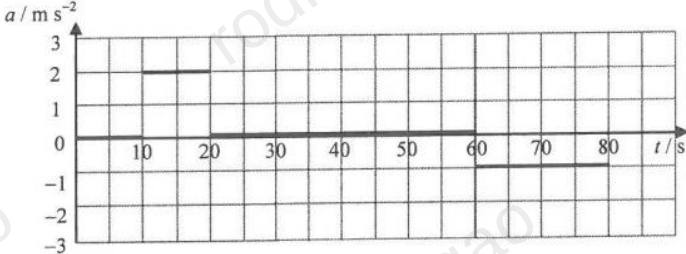


- (a) Describe A's motion throughout $t = 0-80$ s. (2 marks)
- (b)(i) Which car attains the greatest acceleration? Find its value. (2 marks)
- (b)(ii) Sketch B's acceleration-time graph from 0 to 80 s. (2 marks)



- (c)(i) Find the separation at $t = 20$ s. (2 marks)
- (c)(ii) Deduce when B catches A. (2 marks)
- (d) The cars have similar size and shape. Each experiences total resistive force proportional to the square of its velocity. Find the engine-power ratio A:B during $t = 20-60$ s. (2 marks)

Official marking scheme

Solution		Marks	Remarks
3.	(a) 0 – 10 s: constant / uniform acceleration 10 – 80 s: constant velocity / uniform motion	1A 1A	
		2	
(b) (i)	Car B. It is represented by the steepest slope between 10 s and 20 s for B. $a = \frac{20-0}{20-10} = 2 \text{ m s}^{-2}$	1A 1A	
		2	
(ii)			
			
Uniform acceleration between 10 s and 20 s & uniform deceleration between 60 s and 80 s.		1A	
Or Either acceleration or deceleration correct.		1A	
All correct.		1A	
		2	
(c) (i)	Total area of A by 20 s = $\frac{(10+20)}{2} \times 15 = 225 \text{ m}$ Total area of B by 20 s = $\frac{(10 \times 20)}{2} = 100 \text{ m}$ Separation of A & B, $s_{AB} = 225 \text{ m} - 100 \text{ m} = 125 \text{ m}$	1M 1A	
		2	
(ii)	Let B meet A at time $(20 + T) \text{ s}$, $(v_B - v_A) \times T = s_{AB}$ $(20 \text{ m s}^{-1} - 15 \text{ m s}^{-1}) \times T = 125 \text{ m}$ $T = 25 \text{ s}$ $\therefore \text{B catches up with A at } t = (20 + 25) \text{ s} = 45 \text{ s}.$	1M 1A	
	Or $s_A + v_A \times T = s_B + v_B \times T$ $225 \text{ m} + 15 \text{ m s}^{-1} \times T = 100 \text{ m} + 20 \text{ m s}^{-1} \times T$	1M	
(d)	Or Accept using graphical method and read the answer from the graph	1M 1A	
		2	
(d)	Let F_A and F_B be the driving forces developed by the engines of A and B respectively. Ratio of power developed by cars A and B, $P_A : P_B = F_A \times v_A : F_B \times v_B$ $= v_A^2 \times v_A : v_B^2 \times v_B$ $= 3^3 : 4^3 = 27 : 64$	1M 1A	
		2	

(a–b) Velocity–time slope is acceleration, not velocity. A has $a = 15/10 = 1.5 \text{ m s}^{-2}$ initially, then zero. B's four slopes are 0, 2, 0 and -1 m s^{-2} . The original marking graph shows the required horizontal segments.

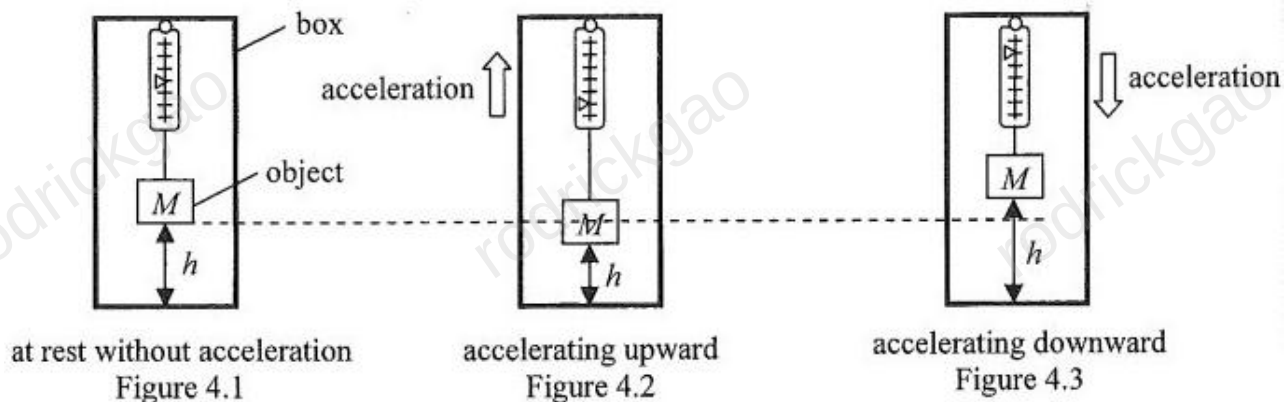
(c)(i) Displacement is signed area under the velocity graph. A travels $75+150 = 225 \text{ m}$ by 20 s, while B travels 100 m, so A leads by 125 m.

(c)(ii) After 20 s B closes the gap at $20-15 = 5 \text{ m s}^{-1}$. Closing takes 25 additional seconds, so the time measured from the original start is 45 s, not 25 s.

(d) Both cars move at constant speed in this interval, so driving force balances resistance. For the assumed common drag coefficient, power is proportional to v^3 , not v^2 .

Question 4

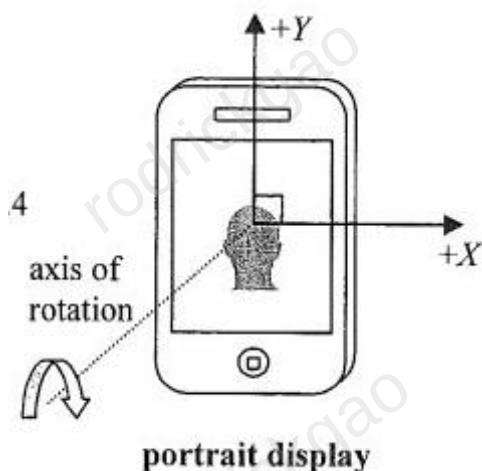
An accelerometer measures acceleration. A mass M hangs from a spring balance inside a box, at height h above its bottom. With no acceleration, the box is at rest (Figure 4.1). Upward acceleration decreases h (Figure 4.2); downward acceleration increases h (Figure 4.3). Spring tension is proportional to extension, so h indicates acceleration.

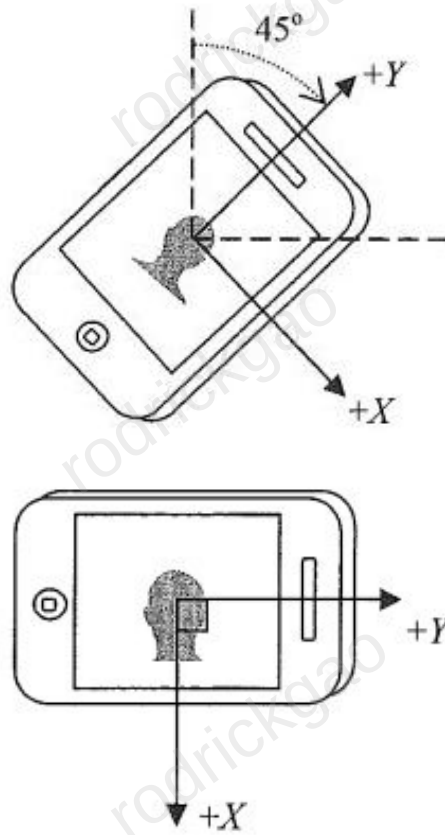


(a) Draw a labelled free-body diagram for M when the box accelerates upwards at a . Explain why h decreases. (4 marks)

(b) The pointer moves 1 cm for every 2 N change of force. The object weighs 5 N. If h decreases by 0.5 cm from its value at rest, find the balance reading and the acceleration magnitude. Take $g = 9.81 \text{ m s}^{-2}$. (3 marks)

Smartphone accelerometers detect gravity along mutually perpendicular phone axes. In portrait orientation, the Y-axis reading is $a_Y = -g$. Rotating clockwise about the horizontal axis perpendicular to X and Y by more than 45° switches to landscape.

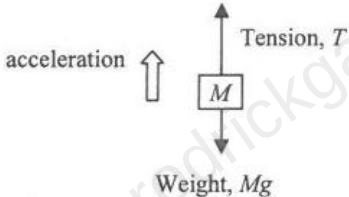




landscape display

(c) What display results when clockwise rotation gives $a_Y = -0.5g$? Explain. (2 marks)

Official marking scheme

Solution	Marks	Remarks
4. (a) Arrows correct	1A	
Labels (tension / T , weight / Mg)	1A	
Or Either force correct and with label	1A	
All correct.	1A	
 As M accelerates upward, $T - Mg = Ma$ i.e. $T = Mg + Ma$ Greater tension needed ($T > \text{original tension} = Mg$) and thus the spring extends further. i.e. M is closer to the bottom, smaller h in Figure 4.2	1M	
	1A	
	4	
(b) Change in tension / reading = $2 \text{ N cm}^{-1} \times 0.5 \text{ cm} = 1 \text{ N}$ $\therefore T - Mg = 1 \text{ N}$ i.e. $T = 6 \text{ N}$	1A	
$T - Mg = Ma$		
$1 \text{ N} = \frac{5 \text{ N}}{g} a$ (accept $M = 0.5 \text{ kg}$)	1M	
$a = \frac{1}{5}g$ or $0.2 g$ (upward) (or $\frac{g}{5} = 2 \text{ m s}^{-2}$ or 1.96 m s^{-2})	1A	
	3	
(c) $a_Y = -0.5g = -g \cos \theta$ $\therefore \theta = 60^\circ > 45^\circ$, 'Landscape display' results.	1M 1A	
Or When the phone is rotated 45° clockwise, $a_Y = -g \cos 45^\circ$ or $-g \sin 45^\circ$ $= -\frac{g}{\sqrt{2}} = -0.71 g$ or -6.94 m s^{-2} (or -7.07 m s^{-2}) 'Landscape display' results as $a_Y = -0.5 g > -0.71 g$.	1M 1A	
Or 'Landscape display' results as magnitude of $a_Y < \text{that at } 45^\circ$	1A	
	2	

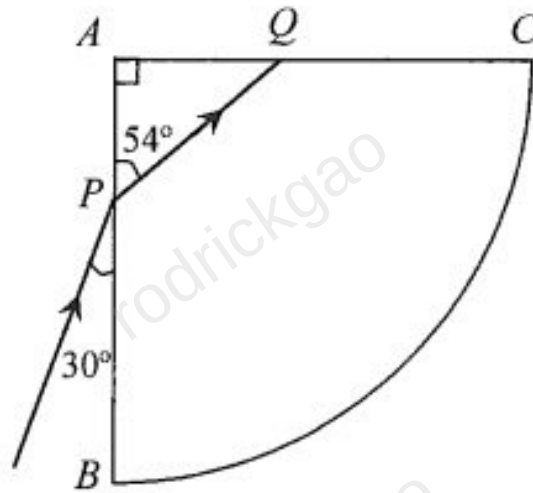
(a) Only tension and weight act on M . Acceleration is not an additional force. Upward resultant force requires $T > Mg$, increasing spring extension and lowering h relative to the box.

(b) A 0.5 cm reduction in h means 0.5 cm more extension, hence 1 N more tension than the 5 N static reading. The balance reads 6 N , but the net force is only 1 N . Mass is $5/9.81 \text{ kg}$, not 5 kg , giving $a = 1/(5/9.81) = 1.962 \text{ m s}^{-2}$.

(c) Following the question's stated sign convention, the gravity component along the rotated Y -axis is $-g \cos \theta$. Its magnitude falls to $0.5g$ at 60° , beyond the 45° changeover. Compare components, not the full magnitude g .

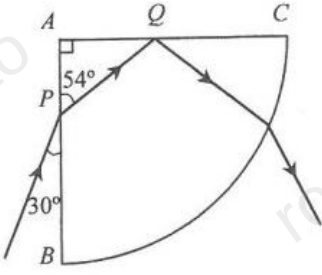
Question 5

ABC is the quarter-circle cross-section of a glass block, centred at A. A red ray enters face AB at P and reaches AC at Q, as shown.



- Calculate the refractive index for red light. (2 marks)
- Explain why the ray is totally reflected at Q. (2 marks)
- Sketch the subsequent path until the ray emerges into air. (2 marks)
- What is observed on emergence if the incident ray is white light? (1 mark)

Official marking scheme

Solution	Marks	Remarks
5. (a) $n = \frac{\sin i}{\sin r}$ (or $n_1 \sin \theta_1 = n_2 \sin \theta_2$) $= \frac{\sin(90^\circ - 30^\circ)}{\sin(90^\circ - 54^\circ)} = \frac{\sin 60^\circ}{\sin 36^\circ}$ $= 1.47$	1M 1A 2	Accept 1.47 to 1.50
(b) $\sin c = \frac{1}{n} = \frac{1}{1.47}$ $c = 42.7^\circ$ (for $n = 1.50$, $c = 41.8^\circ$) Because incident angle at face AC ($= 54^\circ$) $> c$ ($= 42.7^\circ$).	1M 1M 2	
(c)  Reflected ray correct, $i = r$ Emergent ray away from normal.	1A 1A 2	
(d) A spectrum is seen. Or splitting into different colours occurs.	1A 1A 1	

(a) Snell's law uses angles measured from the normal. At AB, $i = 90^\circ - 30^\circ = 60^\circ$ and $r = 90^\circ - 54^\circ = 36^\circ$, so $n \approx 1.473$.

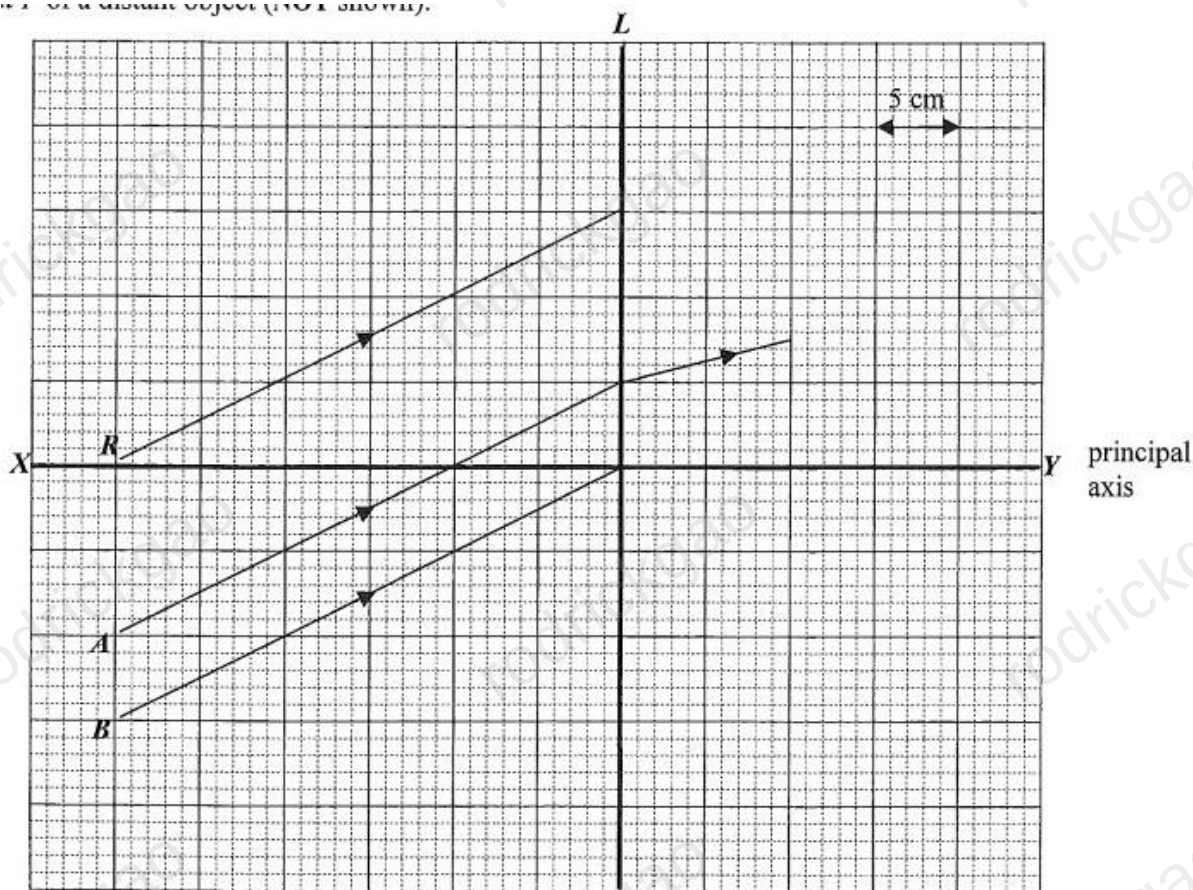
(b) At horizontal AC the normal is vertical, so incidence is 54° . Light travels from glass towards air, and 54° exceeds the critical angle about 42.7° : both conditions for total internal reflection hold.

(c) Reflect downwards and to the right at Q, with equal angles about the vertical normal. At the curved boundary, the normal is the radius from A to the exit point. The ray emerges bending away from that local normal, as in the original marking diagram; it is not automatically normal to the curved surface.

(d) The refractive index depends on wavelength, so white light separates into different colours (dispersion).

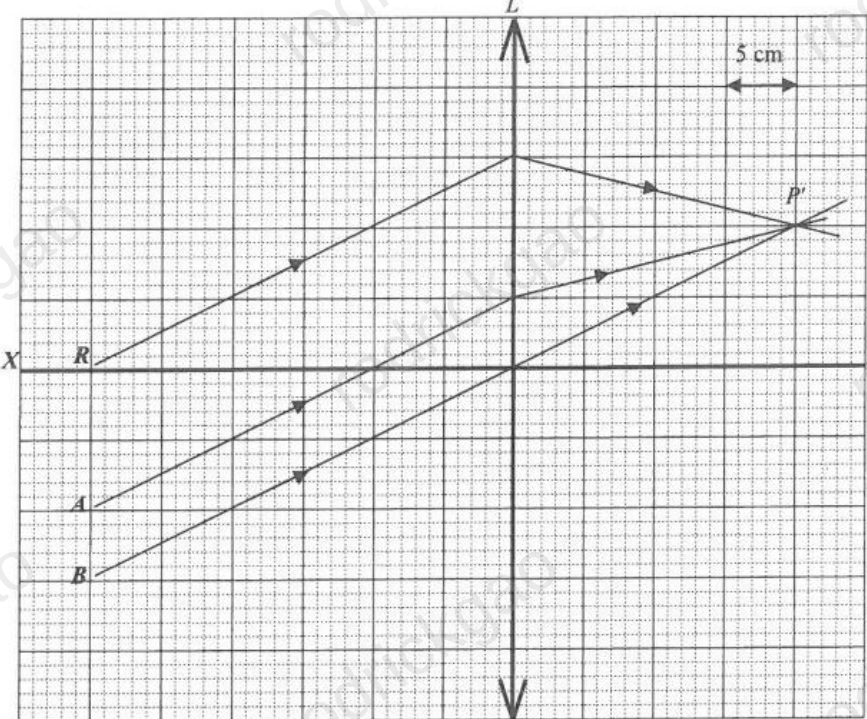
Question 6

In Figure 6.1, XY is the principal axis of thin spherical lens L. A and B are parallel rays from the same point P of a distant object, not shown. Ray R also comes from P. The horizontal scale is shown.



- (a) Identify the type of lens L and explain. (2 marks)
- (b)(i) Locate the image P' of P. (2 marks)
- (b)(ii) Hence determine the focal length. (1 mark)
- (c) Complete ray R after it passes through the lens. (1 mark)
- (d) Describe a simple experiment based on this situation to determine the focal length. (2 marks)

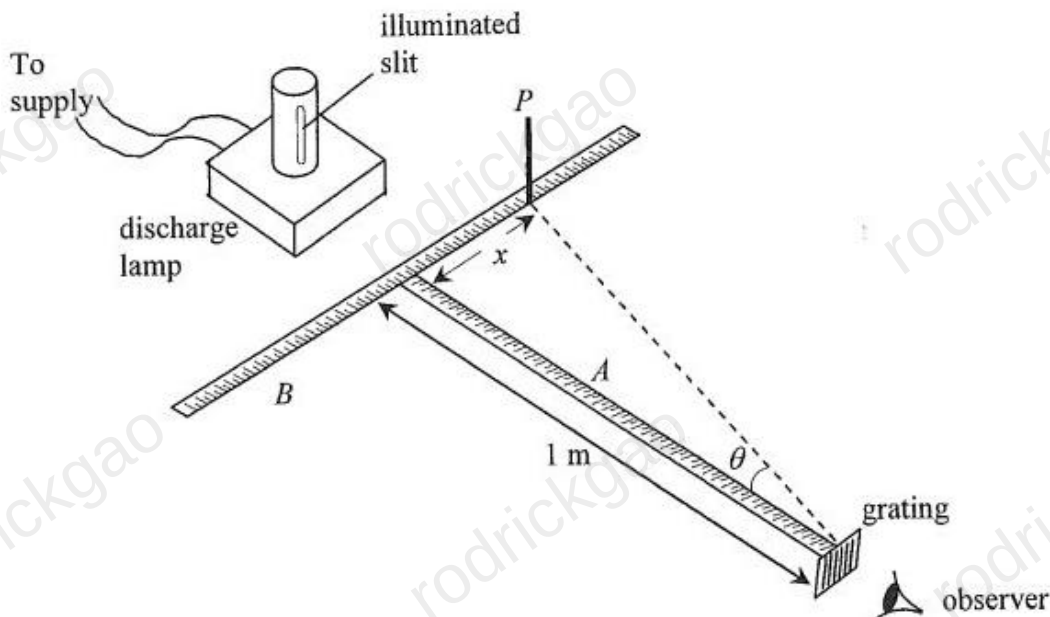
Official marking scheme

Solution	Marks	Remarks
6. (a) Convex lens / Converging lens. Refracted ray of A after passing through L bends towards the principal axis (or optical centre) / converges / bends inward / bends downward. <u>Or</u> A real / an inverted image can be formed. <u>Or</u> The object and the image are on opposite sides of the lens.	1A 1A 1A 1A	
	2	
(b) (i)		
 Rays A and B correctly completed. P' correctly located.		
(ii) $f = 20$ cm	1M 1M 2	
(c) Ray R correctly completed.	1A 1	Accept 19 - 21 cm
(d) Use a screen to capture a/the (sharp) image (of a distant object). The distance between the screen and the lens is f .	1A 1A 2	

- (a) The incident parallel bundle is bent towards convergence. Ray A need not cross the principal axis immediately to show that the lens converges light.
- (b) B passes through the optical centre, so continue it undeviated. Extend the already-refracted A to meet B at P'. The distant source makes P' lie in the focal plane, not necessarily on the principal axis. Measure the horizontal lens-to-image-plane distance: four marked 5 cm intervals give 20 cm.
- (c) All three rays originate at the same point P, so R must refract through P' too. The official original diagram shows the completed rays.
- (d) Aim at a distant object and move a screen until the image is sharp. Measure perpendicular from the lens to the screen; with u much greater than f, $1/f = 1/u + 1/v$ gives $v \approx f$.

Question 7

Two mutually perpendicular metre rules A and B are placed on a bench, with A pointing towards a vertical narrow slit illuminated by a discharge lamp. A grating with vertical lines is at one end of A. A vertical pin P moves along B until it aligns with the second-order image seen through the grating. The distance x is measured as shown.



The grating has 300 lines per mm; the second-order image gives $x = 0.38$ m. Rule A provides the 1 m distance shown.

(a)(i) Calculate θ . (1 mark)

*(a)(ii) Hence find the wavelength of the monochromatic light. (3 marks)

(a)(iii) State one advantage of measuring the second-order instead of first-order position. (1 mark)

(b) The illuminated slit may not be accurately aligned along A. Suggest how to reduce this error. (2 marks)

Official marking scheme

Solution	Marks	Remarks
7. (a) (i) $\tan \theta = 0.38$ $\theta = 20.8^\circ$	1A	Accept 20.8° to 21°
	1	
(ii) $d \sin \theta = n\lambda$ As $d = (\frac{1}{300} \times 10^{-3})$, $(\frac{1}{300} \times 10^{-3}) \times \sin 20.8^\circ = 2\lambda$ $\lambda = 5.92 \times 10^{-7} \text{ m (or } 592 \text{ nm)}$	1M 1M 1A	Accept $5.90 \times 10^{-7} \text{ m}$ to $5.97 \times 10^{-7} \text{ m}$
	3	
(iii) Smaller percentage error in x / the diffraction angle θ . Or Larger x , percentage error decreases.	1A 1A	
	1	
(b) Repeat the procedures with the pin on the left-hand side / the other side of the observer. Take the average value of x obtained from both sides to calculate λ .	1A 1A	
	2	

(a)(i) The triangle gives $\tan \theta = x/1 \text{ m} = 0.38$. Use arctan, not $\sin^{-1}(0.38)$.

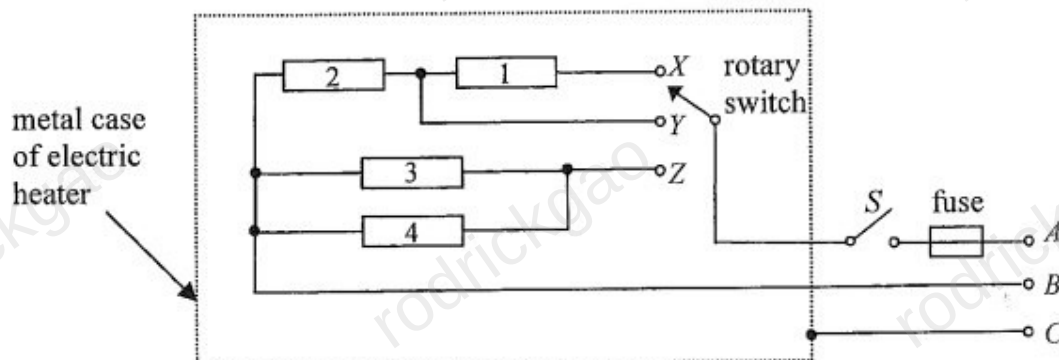
(a)(ii) Convert line density: $300 \text{ mm}^{-1} = 300\,000 \text{ m}^{-1}$, so slit spacing $d = 3.333 \times 10^{-6} \text{ m}$. For the second order, $\lambda = d \sin \theta / 2 \approx 5.92 \times 10^{-7} \text{ m}$. The order number divides the result.

(a)(iii) A similar absolute position-reading uncertainty is a smaller fraction of the larger second-order displacement.

(b) Locate the same order on both sides and average the two distances from the reference. This reduces the effect of a small alignment offset; repeatedly measuring just one side would preserve that systematic offset.

Question 8

An electric heater contains four identical elements, each rated 500 W, 220 V. A rotary switch selects mode X, Y or Z. Wires A, B and C connect to 220 V a.c. mains through a three-pin plug.



- (a) Find the resistance R of one element. (1 mark)
- (b) Find total power in mode X, assuming each element's resistance stays constant. (2 marks)
- (c) Without calculation, identify and explain which mode gives the greatest total power. (2 marks)
- (d)(i) Choose the most suitable fuse from 3 A, 5 A and 13 A. Show your working. (3 marks)
- (d)(ii) A student says that because the supply is a.c., switch S may be in either A or B. Comment. (2 marks)
- (d)(iii) If the live wire touches the metal case, which wire A, B or C prevents a person touching the case from receiving an electric shock? Explain. (2 marks)

Official marking scheme

Solution	Marks	Remarks
8. (a) $P = \frac{V^2}{R}$ $500 = \frac{220^2}{R}$ $R = 96.8 \Omega$	1A 1	Accept 97 Ω
(b) As the total resistance of the circuit for mode X is doubled, total power dissipation $= \frac{V^2}{2R}$ $= \frac{220^2}{2 \times 96.8} = 250 \text{ W}$	1M 1A	Accept 249 W for 97 Ω
<i>Alternative method:</i> As the voltage across each heating element (1 and 2) is halved, power dissipated in each of the heating element $P_1 \text{ or } P_2 = \frac{500}{4} = 125 \text{ W}$ (since $P \propto V^2$)	1M	
Or $P_1 \text{ or } P_2 = \frac{V^2}{R} = \frac{110^2}{96.8} = 125 \text{ W}$	1M	
Or $i = \frac{V}{R_1 + R_2} = \frac{220}{2 \times 96.8} = 1.14 \text{ A}$ $P_1 \text{ or } P_2 = i^2 R = 1.14^2 \times 96.8 = 125 \text{ W}$ Total power dissipation $= 2 \times 125 \text{ W} = 250 \text{ W}$	1M 1A	
(c) In mode Z, the equivalent resistance of the heating elements is the least as they are connected in parallel, hence, under the same voltage, the total power dissipation is the largest since $P = \frac{V^2}{R}$.	2 1A 1A	
(d) (i) For mode Z, the total power dissipated $= 500 + 500 = 1000 \text{ W}$ $I_z = \frac{P}{V} = \frac{1000}{220} = 4.55 \text{ A}$	1M+1M	
<i>Alternative method:</i> $R_{eq} = \frac{96.8}{2} \Omega = 48.4 \Omega$ $I = \frac{220 \text{ V}}{48.4 \Omega} = 4.55 \text{ A}$	1M+1M	
Most suitable value of fuse $= 5 \text{ A}$	1A	
(ii) Although the heater still works in either connection, it is dangerous for switch S to be fitted in wire B (neutral) as the heater / cable would still be live even when the switch was turned off.	1A 1A 1+1	
(iii) Wire C (Earth). Current would be conducted from the case through this wire to the Earth.	2 1A 1A	

(a–b) The 500 W rating applies only at 220 V across a single element. X connects elements 1 and 2 in series; each gets 110 V and dissipates 125 W, for 250 W total, not 1000 W.

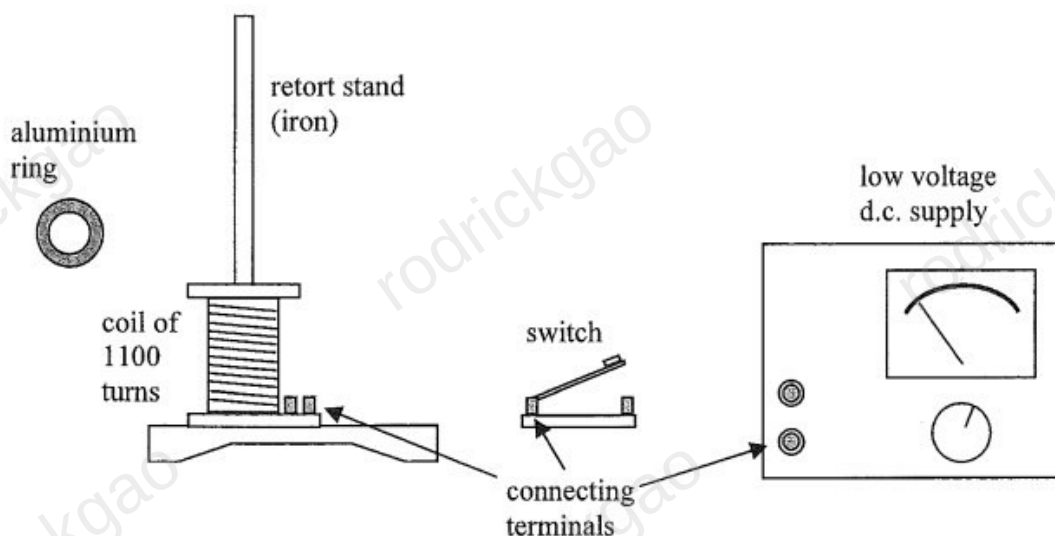
(c) Y connects element 2 alone; Z connects 3 and 4 in parallel, each across the full supply. Equivalent resistances are $2R$, R and $R/2$ respectively.

(d)(i) The fuse must tolerate the highest normal current but interrupt excessive current. 3 A is below 4.55 A; 5 A is the smallest available rating above it; 13 A is unnecessarily high.

(d)(ii–iii) A is live, B neutral and C earth. Alternating current direction does not make live and neutral equally safe relative to earth. The switch belongs in live. In the stated fault, protective earthing provides a low-resistance fault path and allows the fuse to disconnect the supply. This is a circuit-analysis explanation, not a procedure for working on mains equipment.

Question 9

(a) You are given a low-voltage d.c. supply, an aluminium ring, a switch, a coil of 1100 turns and an iron retort stand arranged as shown. Complete the connections using three leads. Describe how to demonstrate Lenz's law of electromagnetic induction; state and explain the observations. (6 marks)



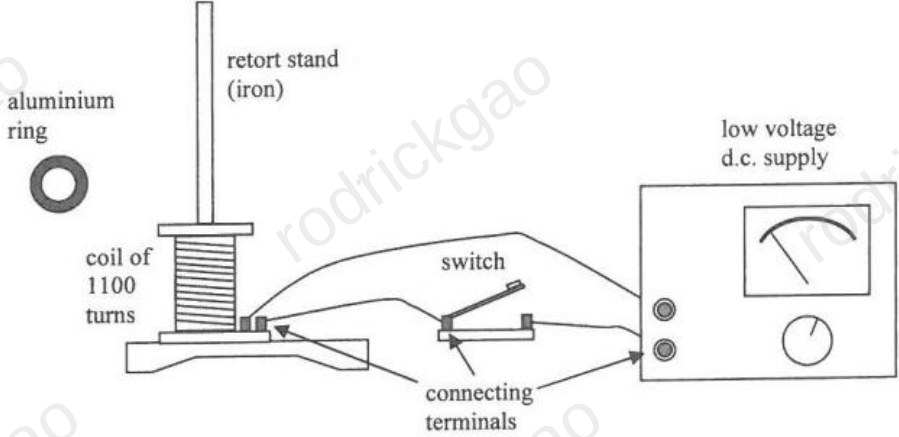
(b) Describe what is observed if the experiment in (a) is repeated with:

(i) a low-voltage a.c. supply; (1 mark)

(ii) a low-voltage a.c. supply and an aluminium ring with a slit through it, as shown. (1 mark)



Official marking scheme

Solution	Marks	Remarks
9. (a) Connect the coil to the terminals of the d.c. supply via the switch (diagram).	1A	
 Put the aluminum ring on the top of the coil through the rod of the retort stand. When closing the switch, the ring would shoot up the rod once, as the aluminum ring experiences a changing magnetic field produced by the coil at the start, According to Lenz's Law, eddy currents flow in the ring to oppose the change. <u>Or</u> A magnetic field in the opposite direction generated by the eddy currents repels the magnetic field produced by the coil. However, when the current and thus the resulting magnetic field are constant, the ring would fall back to the coil as eddy currents no longer flow. (b) (i) The aluminum ring would float in the air. (ii) The aluminum ring with a slit would remain stationary.		
	1A	
	1A	
	1A	
	1A	
	1A	
	1A	
	6	
	1A	
	1	
	1A	
	1	

(a) The three leads form one series circuit: supply → switch → coil → supply. The ring is not wired to the supply; it acts as a separate closed conducting loop. The original marking diagram shows the exact connections.

The growing flux at switch-on induces an emf and current in the ring. Its field opposes the increase, producing the initial upward push. Lenz's law opposes the change in flux, not every magnetic field. A steady d.c. field therefore cannot sustain the induced current or levitation.

(b)(i) With a.c., flux keeps changing. In the stated apparatus the resulting average upward magnetic force supports the ring. Do not claim that the force must be repulsive at every instant merely because the supply is alternating.

(b)(ii) The slit interrupts the conducting path around the ring. An induced emf may still exist, but the large circulating current required for levitation cannot flow; the ring stays put.

Question 10

NASA's Voyager 1 space probe can operate in space for over a decade. Its radioisotope thermoelectric generator (RTG) converts energy released by radioactive decay into electrical energy. The source is plutonium-238, which undergoes α decay.

(a) The source is sealed in a thin metal casing. Explain why the NASA worker shown handling the RTG with bare hands need not be concerned about the emitted α particles. (1 mark)



At launch, the source contained 3.2×10^{25} plutonium-238 atoms. Its half-life is 87.74 years. Take 1 year = 3.16×10^7 s.

*(b)(i) Calculate the activity at launch, in Bq. (3 marks)

(b)(ii) Each decay releases 5.5 MeV. Calculate the source's power at launch, in kW. (2 marks)

*(b)(iii) The question places Voyager 1's departure from the solar system in September 2013, 36 years after launch. Estimate the source's power then as a percentage of its initial power. (2 marks)

Official marking scheme

Solution	Marks	Remarks
10. (a) Alpha particles emitted can be stopped by the (thin) metallic casing.	1A	
Or Shorter range / Lower penetrating power.	1A	
	1	
(b) (i) $k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{87.74 \times 3.16 \times 10^7}$ $= 2.5 \times 10^{-10} \text{ s}^{-1} \text{ or } 7.9 \times 10^{-3} \text{ year}^{-1}$ Activity $A = kN$ $= \frac{\ln 2}{87.74 \times 3.16 \times 10^7} (3.2 \times 10^{25})$ $= 8.000 \times 10^{15} \text{ (Bq)}$	1M 1M 1A	
	3	
(ii) Power = Energy per decay $\times$ Activity $= 5.5 \text{ MeV} \times 8.000 \times 10^{15} \text{ Bq}$ $= 5.5 \times 10^6 \times 1.60 \times 10^{-19} \times 8.000 \times 10^{15}$ $= 7040 \text{ W or } 7.040 \text{ (kW)}$	1M 1A	
	2	
(iii) Power $\propto$ Activity Activity $\propto N$ $\therefore$ Percentage of power left $= \left(\frac{1}{2}\right)^{t/t_{1/2}} \times 100\%$ $= \left(\frac{1}{2}\right)^{36/87.74} \times 100\%$ $= 75.25\% \approx 75\%$ Or Percentage / fraction of power left = 3/4	1M 1A	
Alternatively: $N = N_0 e^{-kt}$ $\therefore$ Percentage of power left $= e^{-kt} \times 100\%$ $= e^{-(\ln 2 + 87.74) \times 36} \times 100\%$ $= e^{-0.2844} \times 100\%$ $= 75.25\% \approx 75\%$	1M 1A	
	2	

(a) This answer concerns external α radiation from the specified sealed source. Low penetration does not imply that radioactive material is harmless if inhaled or swallowed.

(b)(i) Bq means decays per second, so convert the half-life to seconds before finding k . Multiply k by the number of undecayed atoms, not by their mass.

(b)(ii) First convert MeV per decay into joules per decay: $5.5 \text{ MeV} = 8.8 \times 10^{-13} \text{ J}$. Multiplying by decays per second gives joules per second. The 7.04 kW is radioactive thermal power, not the RTG's electrical output; no conversion efficiency is given.

(b)(iii) Each decay releases the same energy, so source power falls in the same ratio as activity and N . Use 36/87.74 half-lives, giving about 75.25% remaining. This is exponential decay, not a linear 36/87.74 reduction.